

Exercises from Modern Cosmology

Abstract

Contents

1	Exercise 1	2
2	Exercise 2	3
3	Exercise 3	5
3.1	4	6

1 Exercise 1

Suppose (incorrectly) that H scales as temperature squared all the way back until the time when the temperature of the universe was $10^{19}\text{GeV}/k_B$ (i.e., suppose the universe was radiation dominated all the way back to *Planck time*). Also suppose that today the dark energy is in the form of a cosmological constant Λ , such that ρ_Λ today is equal to $0.7\rho_{\text{cr}}$ and ρ_Λ remains constant through the history of the universe. What was $\rho_\Lambda/(3H^2/8\pi G)$ back then?

In this exercise we need to find the (energy) density ρ_Λ at time $10^{19}\text{GeV}/k_B$. Therefore, we need a way to related time to temperature and derive the Hubble's rate time dependence. First of all,

$$T_{\text{old}} = 10^{19} \frac{\text{GeV}}{k_B} = 1.1605 \times 10^{32} \text{K}$$

Since the (energy) density is fixed at all times, we have that

$$0.7\rho_{\text{cr}} = \rho_{\text{today},\Lambda} = \rho_\Lambda(t) = 1.316h^2 \times 10^{-29} \text{gcm}^{-3}$$

Therefore, we need to check how H changes with time. Remembering that in a radiation dominated era $a \propto t^{\frac{1}{2}}$, then

$$H \equiv \frac{\frac{da}{dt}}{a} \propto \frac{1}{2}t^{-\frac{1}{4}},$$

We still don't know how to associate a temperature with a time. Through the scale factor, however, we know that the temperature of a thermal bath of photons is

$$T(t) = \frac{T_0}{a(t)} = T_0 t^{-\frac{1}{2}} \implies t = \sqrt{\frac{T_0}{T}}$$

Since we are at it, notice that we do indeed retrieve that H scales as T^2 , since

$$H \propto \frac{1}{2}t^{-\frac{1}{4}} = \frac{1}{2} \left(\frac{T}{T_0} \right)^2$$

The time at the temperature T_{old} is $t_{\text{old}} = 1.523 \cdot 10^{-16} \text{s}$ since the Big Bang. Plugging the numbers, we get the result:

$$\frac{\rho_\Lambda(t)}{\frac{3}{8}H^2(t_{\text{old}})} = \frac{\rho_\Lambda^{(0)}}{\frac{3}{32}t_{\text{old}}^{-\frac{1}{8}}} = 7.67 \cdot 10^{-31} \text{g cm}^{-3} \text{s}^{-2}$$

2 Exercise 2

Assume the universe today is flat with both matter and a cosmological constant, the latter with energy density which remains constant with time. Integrate the equation

$$dt = H_0^{-1} \frac{da}{a} \left[\Omega_\Lambda + \frac{1 - \Omega_\Lambda}{a^3} \right]^{-\frac{1}{2}}$$

where Ω_Λ is the ratio of energy density in the cosmological constant to the critical density. Integrate from $a = 0$ (when $t = 0$) until today at $a = 1$ to get the age of the universe today. In both cases below the integral can be done analitically.

1. First do the integral in the case when $\Omega_\Lambda = 0$
2. Now do the integral in the case when $\Omega_\Lambda = 0.7$.

Hint: Define a new integration variable $x \equiv \ln(a^{-3})$ and then use the fact that

$$\int \frac{1}{\sqrt{1 + \alpha e^x}} dx = -2 \coth^{-1}(\sqrt{\alpha e^x + 1})$$

For fixed H_0 , which universe is older?

Let's do this integral,

$$\int_0^{t_{\text{today}}} dt = H_0^{-1} \int_0^1 \frac{1}{a \sqrt{\Omega_\Lambda + \frac{1 - \Omega_\Lambda}{a^3}}} da$$

We run into trouble! The integral looks mighty for any value of Ω_Λ which is not 0. Luckily, we have the mathematics. By case, we have

Case $\Omega_\Lambda = 0$

The integral reduces to $\int_0^1 \sqrt{a} da = \frac{2}{3}$. Therefore, we get that the age of a flat universe with only matter is

$$t_{\text{today}}^{0,M} = 1.358 \cdot 10^{10} \text{ yrs}$$

Case $\Omega_\Lambda = 0.7$

We need to solve the integral for a positive Ω_Λ . Using the hint we have hint

$$\begin{cases} x \equiv \ln(a^{-3}) \\ dx = -3a^{-1} da \end{cases}$$

Therefore,

$$\int_0^1 \frac{1}{a\sqrt{\Omega_\Lambda + \frac{1-\Omega_\Lambda}{a^3}}} da = -\frac{1}{3\sqrt{\Omega_\Lambda}} \int_0^1 \frac{1}{\sqrt{1 + \frac{1-\Omega_\Lambda}{\Omega_\Lambda} e^x}} dx$$

$$\frac{2}{3\sqrt{\Omega_\Lambda}} \coth^{-1} \left(\sqrt{1 + \frac{1-\Omega_\Lambda}{\Omega_\Lambda} e^x} \right) \Big|_{-\infty}^0 = \frac{2}{3\sqrt{\Omega_\Lambda}} \coth^{-1} \left(\sqrt{1 + \frac{1-\Omega_\Lambda}{\Omega_\Lambda}} \right)$$

Putting it all together, we have that a no curvature universe with matter and a cosmological constant is

$$t = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \coth^{-1} \left(\sqrt{\frac{1}{\Omega_\Lambda}} \right) \text{ old}$$

It doesn't seem good that the result for $\Omega_\Lambda \rightarrow 0$ is not the one from before
For $\Omega_\Lambda = 0.7$ we get

t

3 Exercise 3

Using the fact that the reduced mass of the electron-nucleus in the D atom is larger than in hydrogen, and the fact that the Lyman α ($n = 1 \rightarrow n = 2$) transition in H has a wavelength $1215.67\text{\AA}$, find the wavelength of the photon emitted in the corresponding transition in D. Astronomers often define

$$v \equiv c \frac{\Delta\lambda}{\lambda}$$

to characterize the splitting of two nearby lines. What is v for the H – D pair?

Deuterium is a stable isotope of hydrogen. We can ignore the quantum nature of Deuterium and use Bohr's model to calculate the difference in frequency between the two levels. That is, we use

$$\langle\nu\rangle_{if} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

where n_f is the final level and n_i is the initial one, R_D is Rydberg's number for the *Deuterium*. To compute it, we use the relation between Rydberg's constant, where the nucleus is assumed infinitely heavy, and the reduced mass,

$$R_D = \frac{\mu}{m_e} R_\infty = 1.0971 \cdot 10^7 m^{-1}$$

Plugging the values we get

$$\langle\nu\rangle_{12} = 8.23 \cdot 10^6 m^{-1} \implies \lambda_{12}^D = 121.5355 nm$$

This is the wavelength of the photon emitted when an electron goes from the level $n = 2$ to the level $n = 1$. Now we compute the difference between the two wavelengths, which is $\Delta\lambda = 0.315\text{\AA}$. Finally, we get that the v for the H – D pair is

$$v = c \frac{\Delta\lambda}{\lambda} = 0.00026c \sim 80 \frac{\text{km}}{\text{s}}$$

4 Exercise 4

Convert the specific intensity $I_\nu = \frac{4\pi h \frac{\nu^3}{c^2}}{\exp\left\{2\pi \frac{h\nu}{k_B T}\right\} - 1}$ in $\left[\frac{\text{erg}}{\text{cm}^2} \cdot \text{sr} \cdot \text{s} \cdot \text{cm}^{-1}\right]$. Show that the peak of a $2.73K$ blackbody spectrum lies at $\frac{1}{\lambda} = 5\text{cm}^{-1}$.
